

■■ Google Cloud Associate Cloud Engineer – Skills Checklist

1. Cloud Fundamentals

- 1 Understand Cloud Computing (IaaS, PaaS, SaaS).
- 2 Know Regions, Zones, and Resource Hierarchy.
- 3 Understand pricing models (Pay-as-you-go, discounts).
- 4 Learn the shared responsibility model.

2. GCP Core Services

- 1 Compute Engine – create and manage VMs.
- 2 App Engine – deploy applications.
- 3 Cloud Run & Cloud Functions – run containers/serverless code.
- 4 Cloud Storage – manage buckets and files.
- 5 Cloud SQL & Firestore – work with databases.
- 6 VPC, Load Balancer – configure networking.

3. Identity & Access Management (IAM)

- 1 Create and manage IAM roles and policies.
- 2 Work with Service Accounts and permissions.
- 3 Apply the principle of least privilege.

4. Monitoring & Logging

- 1 Use Cloud Monitoring to create alerts.
- 2 Use Cloud Logging to view and analyze logs.
- 3 Diagnose and fix resource issues.

5. Networking Basics

- 1 Understand IPs, DNS, Firewalls, and Subnets.
- 2 Configure VPC networks and routes.
- 3 Use Load Balancers and Cloud CDN.

6. DevOps & Deployment Tools

- 1 Use gcloud CLI and Cloud Shell effectively.
- 2 Automate deployments with Deployment Manager.
- 3 Manage container images in Container Registry.

7. Security & Compliance

- 1 Use KMS for encryption key management.
- 2 Understand encryption at rest and in transit.
- 3 Manage network security policies.

8. Automation & Scripting

- 1 Learn gcloud commands for automation.
- 2 Practice Linux commands.
- 3 Optionally learn Python or Bash scripting.

9. Troubleshooting Skills

- 1 Fix VM, App Engine, and Cloud SQL issues.
- 2 Use logs to find and solve problems.

10. Soft Skills

- 1 Analytical thinking & problem-solving.
- 2 Good documentation and communication skills.

■ Optional (Advanced)

- 1 Learn basics of Kubernetes.
- 2 Understand Terraform for Infrastructure as Code.
- 3 Explore CI/CD with Cloud Build.